

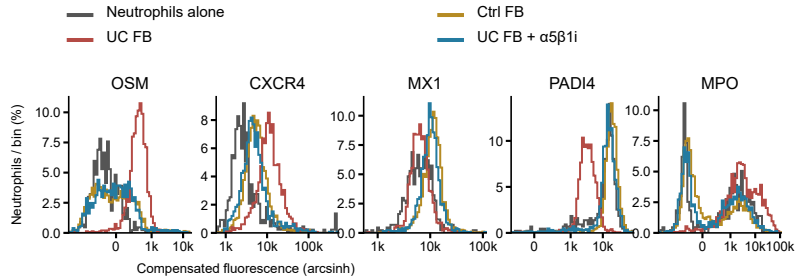
a Culture and blockade design

Peripheral-blood neutrophils

Alone
Untreated / NAMPTi / α5β1i
Ctrl FB
Untreated
UC FB
Untreated / NAMPTi / α5β1i / dual

Independent preculture reference; RNA profiling

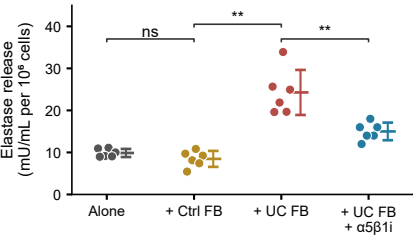
b Neutrophil protein distributions



c Fibroblast exposure and blockade effects on neutrophil proteins

	Fibroblast exposure			Neutrophils alone		Blockade in UC FB coculture				Δ protein z score
	Ctrl FB vs alone	UC FB vs alone	UC FB vs Ctrl FB	NAMPTi vs untreated	α5β1i vs untreated	NAMPTi vs untreated	α5β1i vs untreated	Dual vs untreated	Dual vs NAMPTi	
OSM	+0.07	+1.87**	+1.80**	+0.03	-0.19	-0.01	-1.96**	-1.85**	-1.84*	
CXCR4	+0.83**	+2.86**	+2.03**	-0.04	+0.65**	-1.09*	-2.14**	-2.16**	-1.07**	
MX1	+1.43*	-0.44	-1.87**	+0.47	+1.24*	+0.54	+1.97**	+1.54**	+1.01*	
PADI4	+0.37	-1.75**	-2.11**	+0.13	+0.42	+0.12	+2.32**	+2.07**	+1.95**	
MPO	-0.48*	+2.07*	+2.56**	-0.19	-0.57*	-1.27	-2.13*	-2.18**	-0.91*	

d NETosis Assay (Elastase Release)



e Neutrophil RNA response programs

Neutrophil RNA modules	Ctrl FB vs alone	UC FB vs alone	UC vs Ctrl FB	α5β1i alone	α5β1i in UC FB
Inflammatory cytokines	+0.32*	+0.40***	+0.08	+0.35***	-0.07
Interferon response	+0.06*	+0.04	-0.01	+0.06*	+0.01
Granule/protease	+0.02	+0.02	-0.02	+0.07	+0.06***
NADPH oxidase	+0.25**	+0.28**	-0.02	+0.28***	-0.05
Chemotaxis	+0.46***	+0.48***	-0.00	+0.50***	-0.14*
Hypoxia/glycolysis	+0.19**	+0.14***	-0.06	+0.20***	+0.01

f Predicted fibroblast–neutrophil routes

	UC vs Ctrl FB	α5β1i vs UC FB
1 COL1A2 → CD44 ECM-FB → OSM	-0.79	+0.82
2 FN1 → CD44 ECM-FB → CXCR4	-0.64	+0.67
3 IL1B → IL1R2 CCN-FB → MX1/ISG	-0.45	+0.62
4 CEACAM1 → CEACAM8 ECM-FB → PADI4	+0.41	-0.41
5 CXCL12 → CXCR4 CCN-FB → CXCR4	+0.65	-0.65
6 C5 → C5AR1 ECM-FB → CXCR4	+0.79	-0.79

LR consensus change (−0.9 to +0.9)

g Module support for ligands in f

	Inflam. cytokines	IFN response	Granule/protease	NADPH oxidase	Chemo-taxis	Hypoxia/glycolysis
1 COL1A2	5	6	2	1	1	8
2 FN1	5	10	2	1	1	7
3 IL1B	8	10	3	0	4	2
4 CEACAM1	5	10	2	2	2	7
5 CXCL12	7	10	2	2	1	7
6 C5	8	10	2	1	1	7

Matched ligands; color: prior rank; dot size/count: top decile